Ulta Setting Spray: Pricing, Value & Market Segmentation

Summary

This project analyzes 156 setting spray listings from Ulta (59 brands, 2026 snapshot) to answer a simple question: where is the value, and what are shoppers paying for? Using Python for the analysis and Tableau for visualization, I tested for a small-size price premium, compared finishes, ranked brands into price tiers, and used clustering to find natural market segments.

Key findings

- **Small bottles carry a real premium.** The median Small size costs about $14.41/oz versus about $8.81/oz for Large, roughly 1.6x. The difference is statistically significant (Mann-Whitney, $p < 0.001$), and Small costs more per ounce than Large at 20 of the 22 brands that sell both.
- **Finish does not drive prices.** Dewy, Matte, Natural, Hydrate and Radiant show no significant price difference (Kruskal-Wallis $p = 0.70$), and none is significant after controlling for size tier.
- **Brand is the real price driver.** Budget brands are roughly 3-4x cheaper per ounce than prestige brands.
- **Four market segments emerge.** Value, Everyday, Premium, and a Small / mini sizes cluster that reflects the small-size premium.

Data and preparation

The dataset contains one row per product-size listing, with brand, product name, price, size (oz), price per ounce, finish type, and a size tier.

Size tier rule. One size available: Medium. Three sizes: Small, Medium and Large. Two sizes: Small and Large. A mini: Small.

Cleaning steps

- Removed 4 exact duplicate rows (Charlotte Tilbury and Morphe each had a Small and a Large listed twice), leaving 156 listings.
- Trimmed whitespace and standardized categories (finish and size tier labels).
- Validated that price ÷ size matches the stated $/oz on every row, and confirmed there were no missing values.

Methods

- **Exploratory analysis:** distributions of price, size and $/oz; Spearman correlations.
- **Hypothesis tests:** Kruskal-Wallis across size tiers and finishes; one-sided Mann-Whitney for Small vs. Large (non-parametric because $/oz is right-skewed).
- **Regression:** OLS on log $/oz with finish and size tier, to check whether finish matters after controlling for size.
- **Clustering:** k-means (k = 2 to 7) on standardized log(price) and log($/oz). Price, size and $/oz are mathematically linked, so I avoided using all three. Hierarchical clustering served as a robustness check.
- **Tools:** Python (pandas, SciPy, statsmodels, scikit-learn, seaborn) and Tableau.

Findings

The small-size tax

Price per ounce falls as bottles get larger. Small (including minis) has the highest median $/oz, and the Small vs. Large gap is about 1.6x.

Across the 22 brands that sell both a Small and a Large, 20 charge more per ounce for the Small. The premium is steepest at Morphe (about 2.8x), RMS Beauty (about 2.4x), Charlotte Tilbury (about 2.2x) and Urban Decay (about 2.1x). It is absent at Tarte and Kopari.

Note on the Medium tier: Medium mixes single-size products (often drugstore brands) with the middle size of a three-size ladder, which is why it looks cheapest. Small vs. Large is the cleanest comparison.

Finish

Radiant and Hydrate have somewhat higher median $/oz, but the differences are within noise. Finish does not meaningfully explain price once size tier is accounted for.

Brand price tiers

For the 16 brands with at least three listings, I split median $/oz into terciles (cutoffs about $4.07 and $8.50).

Tier	Example brands
Budget	Essence, L.A. Girl, Revolution, ULTA Collection, e.l.f.
Prestige	MAC, Charlotte Tilbury, Anastasia Beverly Hills, Tarte, Milk Makeup

The gap in median $/oz between budget and prestige brands is roughly 3-4x.

Market segments

K-means with k = 4 produced these segments:

Segment	Listings	Median price	Median $/oz
Value	41	about $10	about $3.85
Everyday	34	about $18	about $5.92
Premium	48	about $36	about $10.61
Small / mini sizes	33	about $18	about $17.78

The fourth segment is defined by bottle size (median about 1.15 oz), not brand tier. It is largely travel and mini sizes from prestige brands, so it is the small-size premium showing up as a cluster. Silhouette is highest at k = 2 (0.52), but k = 4 is nearly as good (0.48) and far more interpretable. K-means and hierarchical clustering agree closely (adjusted Rand index 0.87).

Implications

- **For shoppers:** the best value per ounce is in larger bottles. Paying for a mini is mostly paying for convenience.
- **For brands and merchandisers:** price position is set by brand and size, not by finish. Finish is a positioning or marketing choice rather than a pricing lever.
- **For assortment planning:** a size ladder with a clear per-ounce discount at each step is the norm; brands without that discount (Tarte, Kopari) stand out.

Limitations

- One retailer (Ulta) and one snapshot in 2026; promotions and prices change.
- Small sample (156 listings, 59 brands); many brands have only 1-3 listings.
- Size tiers were assigned by rule rather than measured, and Medium mixes single-size and mid-size products.
- No sales, rating or review data, so "value" here means price per ounce, not quality or popularity.

Next steps

- Pair this shelf-level view with the GMI Insights setting spray market report to compare the finishes and segments that are growing against what Ulta actually stocks.
- Collect a second snapshot to measure promotion and price changes over time.
- Add ratings and review counts to test whether higher price buys higher customer satisfaction.

Deliverables

- Python notebook with the full analysis (Ulta_Setting_Spray_Analysis.ipynb)
- Cleaned and clustered datasets (setting_spray_clean.csv, setting_spray_clustered.csv, setting_spray_tableau_ready.csv)
- Interactive dashboard (published page, with Tableau build instructions)